

"Google Maps can say a lot of things, but it cannot guide us properly": Exploring Technology Inaccessibility for People with Visual Impairments in Bangladesh

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Abstract

Worldwide, people with visual impairments (PVI) face significant barriers in accessing technologies. These barriers exacerbated in low-resource contexts like Bangladesh due to infrastructural limitations. This paper explores the technology usage and preferences of n=7 tertiary-level students with visual impairments in Bangladesh. The study findings show that the technology-related challenges of PVIs include failure of assistive technologies. Participants shared that they face most inaccessibility while using Navigation technologies due to the infrastructural scenario of the country and lack of contextualization of those technologies. This research emphasizes the opportunity for the navigation technologies to better serve PVIs, particularly in regions with similar socio-technological challenges. This work presents the technology challenges and improvement suggestions of the participants, and outlines the need for participatory design research for more accessible and contextually aligned navigation technologies for this community.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**.

Keywords

Visual Impairment, Technology Preferences, South Asia, Low-resource Context

ACM Reference Format:

Eusra Amreen, Mahir Morshed, Ifti Azad Abeer, and Nova Ahmed. 2026. "Google Maps can say a lot of things, but it cannot guide us properly": Exploring Technology Inaccessibility for People with Visual Impairments

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CHI EA '26, Barcelona, Spain

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ACM ISBN 979-8-4007-2281-3/26/04
<https://doi.org/10.1145/3772363.3799021>

in Bangladesh. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3772363.3799021>

1 Introduction

There is limited social and technological support for the communities of people with visual impairment (PVI) worldwide [19, 23]. The structural and social aspects in resource-limited regions further exacerbate challenges of PVI [6, 13, 18]. The technology design and development are also limited for this community in resource-constrained regions like South Asia [6, 13, 18], as can be seen in Bangladesh, a South Asian country where the research is based. A recent study showed the structural barriers impacting educational access such as lack of compassionate and aware teachers, lower number of policies etc. [13, 28]. Navigation is also challenging since the number of routes and pathways that support the PVIs are limited to urban walkways creating mobility problems [1, 15]. The mobility challenges directly impact resource access and this community struggles to access basic and specific healthcare support [14, 31]. Furthermore, affordable public transport is not friendly towards PVIs [15]. There have been reported cases of increased amount of sexual harassment and abuse towards women from the PVI community [14, 26]. These challenges intensify when community and family members consider them as a burden [27]. These challenges create barriers towards various aspects such as receiving education [28] or achieving financial independence [13].

In South Asian countries, with its own sets of challenges, there is a requirement for technology design, development, and deployment to meet the sociocultural issues and infrastructural barriers. Prior research in technology for PVIs has also shown the distinction of South Asian regions compared to global developments. While globally researchers in this domain focused towards bringing out cutting-edge technologies tailored to PVIs emphasizing on specific application areas, the usability improvements and accessibility enhancements [3, 5, 10, 11, 16], research based in South Asia primarily

focused on the state of PVIIs in terms of technology access and usage [2, 17, 18, 24]. Solution approaches have focused on specific services, such as access to digital financial services for this community living in India [17]. The design work focused on supporting the educated PVIIs considers low-cost, locally developed affordable technology [4, 6, 18]. The affordability of smartphones in this region has opened up great opportunities that meet the regional requirements.

The challenges PVIIs face call for technology-enabled solutions and support. However, in terms of technology support and solutions targeted for PVIIs in Bangladesh, initiatives are limited in rigor and implementation. While there is an existing rich body of research to support PVIIs, there is a gap in the exploration in the HCI literature on the tertiary-level students with visual impairments in Bangladesh exploring their technology requirements. This work aims to bring out the current technology accessibility needs of the tertiary-level students with visual impairments living in Bangladesh through the exploration of the following research questions:

RQ1: What are the major accessibility barriers for the students with visual impairments in Bangladesh?

RQ2: Which accessibility barrier needs technology design prioritization for this community?

To answer these research questions, this research considers a qualitative study focusing on seven PVIIs ($F=1$, $M=6$), pursuing tertiary level education over a period of six months. Findings show that the technology space lacked accessibility and contextualization in navigation and educational technologies, banking applications along with basic inaccessibility issues which becomes more complicated in resource-constrained scenario of the country. The participants suggested technology improvements which included navigation, and technology learning support along with education and assistive technology improvements. They shared how navigation technologies such as Google Maps and smart canes failed to address their needs due to the dense urban settings of the country. Participants reported that voice instructions are often misaligned with informal pedestrian routes, obstacle density overwhelms vibration-based feedback, and mapping accuracy does not reflect congested street realities. These contextual mismatches illustrate how infrastructural conditions directly affect the usability of existing tools.

While the study broadly explored technology accessibility barriers experienced by PVIIs, navigation-related technologies emerged that navigation technologies need immediate focus for this community. We present these design preferences and the challenges shared by the participants to guide future research to contribute in this contextualized problem space through human-centered design research.

This poster makes a three-fold contribution to the HCI community. Firstly, it provides empirical insights into technology inaccessibility experienced by PVIIs in Bangladesh. Secondly, it presents how infrastructural constraints and limits the effectiveness of existing navigation tools. Finally, it sheds light on opportunities that HCI researchers have for human-centered design research to bring out contextualized navigation technology for this population.

2 RELATED WORK

Globally, the recent literature regarding the technology usages and preferences of PVIIs focused on enhancing the accessibility and usability of smartphone applications. Researchers explored how PVIIs used speech input compared to on-screen keyboards [3], how they manage accessibility challenges in typical office environments [5], ways to safeguard them from viewing ableist comments [11], introduced and evaluated a scalable method for collecting bus stop location and landmark descriptions [10], and created a personal object recognizer for identification of surrounding objects [16]. These works are specific task-oriented and focus on improving accessibility for specific tasks or platforms. Whereas, the research works in South Asia and Bangladesh, where the research is based, primarily focused on the technology usage scenario and more broader technology suggestions. This research aims to identify which barriers should be prioritized, specifically in the resource-constrained contexts.

Technological Challenges in South Asia: PVIIs have to face many difficulties because of infrastructural and social restrictions. The main challenges were the high cost of devices and apps, limited technical support, and navigational issues [25]. Silva and Prasad discussed the daily challenges faced by PVIIs in Sri Lanka, particularly focusing on the difficulties of independent walking, navigation, and the limitations of current assistive electronic travel aids (e.g. false alarms in obstacle detection and recognition) [34]. Shashank et al. proposed an aid system developed using object detection and depth perception to enable a person to navigate without dashing into an object [20]. They implemented a navigation feature to navigate the PVI to his/her destination. Chakkaravarthy et al. developed MagicEye, a wearable device designed to assist PVIIs to recognize indoor and outdoor objects [32]. Khan and Khusro suggested that RFID tags store location information can be retrieved within a proximity of up to 10–15 m, which, integrated with white canes, can be used for navigation [18]. Apart from navigation, several other challenges and corresponding solutions were identified. PVIIs encounter challenges in input interaction with devices, display interfaces and overall accessibility [22, 33]. Assistive Technologies play a crucial role in resolving the accessibility challenges of PVI where Mahir et al. conducted a study which introduces SPAVIS, a mobile application designed to assist SVIIs with their microtasks [21].

Challenges and Technological Interventions in Bangladesh: In Bangladesh, assistive technologies access remains as a challenge. In this resource-constrained context, social and infrastructural barriers shape everyday technology usage of the PVIIs. According to Hossain et al., back in 2010 among 152.2 million people [8], over 13 million people in Bangladesh were physically impaired, of which 3 million were children. Abir & Hoque presented that there are three main types of barriers (Social, Psychological, and Structural) to accessibility in Bangladesh [1]. Hossen investigated the employment barriers, supports, and outcomes and suggested that employment negative societal attitudes, lack of accessible workplaces, limited education and training, and insufficient employment support systems acted as barriers for the PVIIs [13]. Researchers found that PVI girls are the easiest victims of sexual abuse [26] and reported that existing modes of transportation are inadequate and non-inclusive

and pose both physical problems (such as roads, ramps, footpaths, etc.) and social barriers [15].

Rose & Ghazali conducted a study on the type of design and interaction of a navigation application that are more convenient and favored by PVI in Bangladesh and found that merely traveling with the white cane could not avoid problems faced when going from one place to another [30]. Study results indicated that Voice-assisted devices, layout with more spacing, and audio descriptions were preferred by PVI. Rahman et al. presented a wearable device to detect the obstacles in front of the user, hump on the ground and moving objects, detect the sudden fall, and inform the user's guardian [29]. This system generates audible instructions to navigate the user properly.

In summary, while hitherto research explored task-specific assistive technologies, understanding how everyday accessibility barriers accumulate in resource-constrained context and which barriers should be prioritized for design. We try to address this gap by exploring technology accessibility barriers and preferences among tertiary-level students with visual impairments in Bangladesh.

3 RESEARCH METHOD

We carried out a qualitative study to discover the challenges faced by PVI in Bangladesh. The study is focused on technological challenges, especially with regard to smartphones.

Participants: We conducted semi-structured interviews with seven participants (Male= 6, Female=1) from various regions of Bangladesh to bring out the existing barriers they face in the context of low-resource context and their technology priorities. Participants' details e.g., age, gender, type of disability, region, and educational level, are given below in Table 1 in Appendix. Participants were selected using the snowball sampling method. We initially contacted the participants, shared the study objective over phone call, and asked about any reservations or preferences they had.

Interview Procedure We initially conducted individual interviews with the 7 participants on both WhatsApp and Google Meet, where we asked them about the technological challenges they face daily and their technological preferences. Then, we conducted a focus group interview on a WhatsApp group call with 5 of our initial participants, where we asked them about some of their common challenges and their preference for solutions for those challenges. The language of the interview sessions was Bangla (the native language of participants and researchers). The interviews started with greeting and asking about participants' demographic questions and tried to maintain participants' comfort in every situation in the interview session. One researcher moderated the interview, and others took notes. The moderator used the introduction as an opportunity to describe the topic to the participants and asked about their choices, worries, or any reservations regarding the issue of our interest.

Qualitative Content Analysis: We started the qualitative content analysis by transcribing the audio recording of the interviews. The grounded theory approach was applied in the Coding process [7], aided by the Computer-aided Qualitative Data Analysis Software Taguette. During the open coding phase, we read the transcripts line by line, reflected deeply on the interview contents,

and identified concepts, participants' experiences, and challenges mentioned in the transcripts. We identified 63 initial codes (e.g., Bangla TTS, Talkback Issues, Navigation issues, etc.) in the open coding phase. We continued refining the codes throughout the study. To identify the high-level themes, we started axial coding by categorizing the codes generated in the open coding phase. The codes include but are not limited to technology challenges: access to technology, technology challenges: available resources., and technology use: impact of technology. Two researchers participated in the initial coding phase and agreed on most of the codes. In case of any disagreement to reach a consensus, researchers discussed it among themselves.

Research Ethics and Incentives: The Institutional Review Board (IRB) approved this study. Before the interview, we discussed research aims, principles, interview techniques, guidelines, data anonymity, and permissions with participants. Participants could quit the conversation at any point in the interview session. Interview notes, recordings, and write-ups are securely stored on a secure drive that only the researchers have access to.

Positionality: All of the researchers were born and brought up in Bangladesh. The research was conducted by mixed-gender researchers from different socioeconomic backgrounds with several years of qualitative research experiences in Bangladesh on technology and special groups. One of the authors has the experience of exploring the technology exposure of people with visual impairment in the same context of our study. The researchers' backgrounds, positions, and research exposures have impacted the research lens to conduct the qualitative research [9].

4 FINDINGS

During the interviews, participants shared multiple accessibility barriers they face. These include challenges related to navigation, education, banking, and everyday smartphone interactions. While these challenges ranged across different domains, navigation technology-related ones were reported most frequently which showed the need for prioritization regarding navigation technology. Participants emphasized that these challenges directly affected their access, mobility, and safety.

4.1 Inaccessibility in day-to-day technologies where sociocultural factors worsen things

The participants' technology usage was tamed due to the frequent inaccessibility issues they faced. There are a number of accessibility related challenges mentioned by the participants where almost one-fourth of the accessibility challenges stemmed from Navigation and one-fifth from Mobile Banking (presented in Figure 2 in Appendix). Such distribution of accessibility challenges suggests that navigation technology related barriers dominate the participants' everyday technology frustrations, putting navigation as one of the key design priorities emerging from the interviews.

The geographically incompatible navigation devices meant using those devices generated significant concern among them rather than serving their purpose of easing the navigation task. The infrastructural issues along with the lack of contextual considerations from the manufacturers of those technologies led to many inaccessibility issues. For example, Bangladesh being a congested, densely

populated country with a developing economy, was not suited for certain specifications of the smart canes used by the participants. Participant, P2 shared as follows.

"A smart cane alerts us to obstacles within 3 meters through vibrations. In our country, broken and dirty roads can have many obstacles within a 3-meter range, so there needs to be a solution for this. If there are obstacles all around, it's difficult to understand the exact location just through vibrations. Having an accessible map would be really helpful. Google Maps can say a lot of things, but it cannot guide us properly, so if they could improve their sensor, then it would be helpful." - P2

This quotation indicates how assistive technologies designed for low-obstacle environments may not serve their purpose densely populated areas. Inaccessibility dominated other applications they used also, be that the mobile banking applications or simply a service-oriented application like ride-sharing applications. Some of the mobile banking applications were not compatible with 'Talk-back' (a screen reader that most participants used) making them inaccessible for the participants.

Social discrimination is observed in all spheres of PVI's lives and their digital world interactions are not free from that too. Participant P8 mentioned that they face discrimination in the ride sharing applications. Such negative attitudes towards them peaked even further when it came to monetary transactions as a participant quoted sadly.

"While aware and honest people don't deceive us, many rickshaw drivers overcharge. If the fare is 50 taka, they might take 100. When I ask others about the price of something, some say it's 10 taka, while others say it's 100. There are dishonest people everywhere, and this is a real problem." - P4

Moreover, they have to face numerous social stigmas that sting them in their daily lives and influence many decisions they take or perceptions they are subjected to. Participant P7 mentioned one such aspect by quoting, *"Given my visual impairment, I was initially in the Commerce group since visually impaired individuals often struggle with mathematics and complex technical terms."* Likewise, there are significant barriers that a PVI needs to overcome to receive education. While financial and other applications also posed accessibility challenges, participants frequently described navigation as the most urgent technological limitation that affected their lives which is further illustrated in the next subsection.

4.2 Technology Preferences and Suggestions for Improvements

All of our participants reported Navigation and mobility difficulties. This challenge invoked our participants' preference for the improvement of Google Maps sensors for proper guidance, while another demanded an interactive Google Map that can give you voice feedback on voice commands (illustrated in Figure 1. He quoted as follows.

"Google Maps doesn't have an interactive system which can provide voice feedback on 'where are the nearby

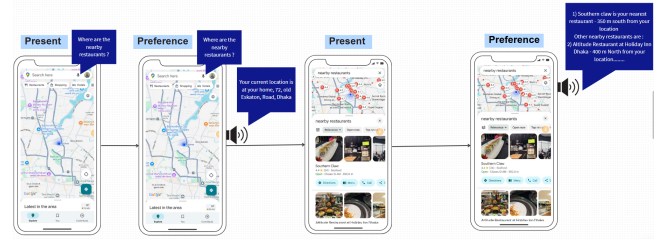


Figure 1: The visual representation of their suggestions for improvement of Google Maps for better navigation and mobility.

restaurants?'; 'How to get to the nearby restaurant?' etc., which is why we are unable to use it." - P2

The accuracy of Google Maps also becomes highly critical in densely populated country like Bangladesh, especially in congested areas where it often fails to navigate properly. Participants P5 wanted them to address this issue and suggested sensor improvements saying, *"Google Maps provides a lot of information, but it can't always guide properly. It would be better if their sensors could be improved."*

4.2.1 Display and Accessibility Enhancements. The participants wanted a few display adjustments along with accessibility enhancements in the existing design of different applications and the operating system in general. Generally, when you press a button on a smartphone, no explanations are provided for its purpose, but it makes it harder for a PVI to utilize the button properly. One of the participants, P1, expressed his preferences as follows.

"It would be better if, when pressing a button in the app, a pop-up appeared explaining what it means. Because other than that it is very difficult for us to know the use of the button and use it properly. Also, I think if a solution can be found to type faster, that would be really great." - P1

P1 was one of the most tech-savvy participants that we had and used paid and licensed version of some of the assistive technologies which may have impacted the type of issues that he has faced. The issues he raised were rather advanced usability issues. Whereas P3 who had low vision wanted adjustments in font size *"It would be great if I had the option to increase the font size even more."*

Thus, these preferences demonstrate the difference in accessibility needs of the PVI's based on their vision type and technological familiarity. This scenario suggests that one-size-fits-all accessibility solutions may not be sufficient and indicates design implication which is detailed in the next section.

5 DESIGN IMPLICATION FOR ACCESSIBLE NAVIGATION AND WAY FORWARD

The study explored the technology accessibility barriers faced by the tertiary-level students with visual impairment in Bangladesh, as a resource-constrained context, where infrastructures lack accessibility and technologies lack contextualization. We interviewed $n = 7$ PVI who are tertiary-level students and critically explored their technology preferences in the light of their current technological

scenarios. Their stories reflected the different barriers that they face regarding technology usage complicated by the infrastructural complexities. Such characteristics of their living environment yield multifaceted discrepancies, where navigation technologies remain the most inaccessible.

Our findings indicate that assistive technologies for navigation failed due to congestion, point towards the area that needs to be addressed through prioritization of contextual norms. While existing application like Google Maps provide voice instructions, it is not focused specifically on PVI. The instructions needs to be more frequent with added functionalities of identifying nearest obstacle, broken roads, crowded areas, and many more context-specific feedback. While smart canes try to address this issue, they also fail due to similar infrastructural challenges. According to the insights of this work, we outline the following specific design implications to be addressed in future explorations.

Adaptive design of navigation technologies for dense urban contexts. Navigation systems should adapt its feedback based on the obstacle frequency, road condition, crowd density, and other accessibility considerations. Hossain et al., tried to address the navigation difficulties for wheelchair users through automated detection of environmental barriers, and step by step detailed navigation where they tried out this approach for two university campuses in the United States [12]. They vouched such approach to address urban accessibility, smart mobility, and to bring out inclusive navigation solutions. Likewise, a navigation technology can be designed for the PVIs of Bangladesh by crowd-sourcing road accessibility information and integrating them into routes of the map.

Involvement of PVIs while designing interactive navigation interfaces. Participants emphasized the need for conversational voice queries and real-time guidance for which they many a times need to take an acquaintance with them while commuting. Specific and personalized needs and variety of challenges faced by the participants call out for a crying need of human-centered design research through participatory design approach by involving the PVIs as design partners beyond stakeholders. Involving PVIs of the specific communities as active design partners is needed to make the navigation technologies more aligned with the contextual demands and lived experiences of the PVIs and to make them better equip to travel and move on their own.

While this exploratory qualitative study is limited to a small sample of tertiary-level students, it highlights how accessibility barriers intensifies in resource-constrained settings and how navigation technologies require prioritization. Future research will look to build on these insights through engagement with advanced artificial intelligence (AI) driven technologies in consideration with above-mentioned design implications by prototyping and evaluating contextualized navigation technology and extending this investigation to broader PVI communities.

Overall, this work contributes to HCI by providing empirical insights into technology inaccessibility and presenting the importance of contextual alignment in assistive technology design. Additionally, it highlights the need to move beyond generic accessibility solutions toward context-specific, personalized, and participatory approaches.

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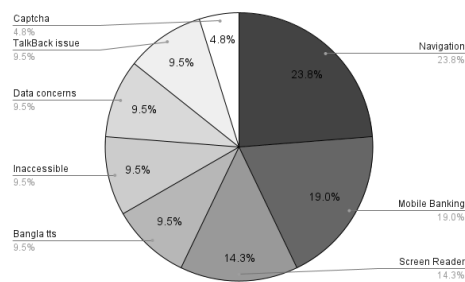


Figure 2: Accessibility challenges percentage among participants

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A Appendix

The table 1 below provide the participants’ details such as age, gender, type of disability, region, and educational level.

Figure 2 presents the types of accessibility challenges and apps that consist of accessibility.

Table 1: Participants' Details

ID	Age	Gender	Cause of visual impairment		Age of onset of Blindness	Region	Education
P1	26	Male	Retinitis	Pigmen- tosa	14	Urban	Studying masters in English
P2	21	Male	Ophthalmia		8	Urban	Completed HSC*
P3	23	Male	Not mentioned		0	Rural	Completed HSC
P4	22	Male	Typhoid		-	Urban	Completed Bachelor
P5	40	Male	Not mentioned		0	Urban	Studies Bachelors in Government and politics
P6	21	Male	Not mentioned		4	Urban	Studying Honours first-year English Literature
P7	31	Female	Fever		15	Urban	Completed Bachelors